CHAPTER 3

REALITY

LATELY, AI RESEARCHERS have devoted serious attention to the project of giving machines "groundedness"—a reliable relationship between the machine's representations and actual reality—as well as memory and an understanding of causation. New technical methods are unlocking improvements in these abilities, and further progress will undoubtedly follow.

All these advances will contribute to the ultimate goal of producing a new genus of AI: machines that can not only interpret our real world but also plan in it. Today's systems, by contrast, output their answers linearly, on the basis of correlation; they cannot internally create a model, or prototype, of their future moves, and they are only beginning to form conceptions of causal relationships. Similarly, today's game-playing AIs can forecast the probable consequences of their moves only within the limited and highly abstract confines of a digital frame.

Planning machines would need to combine the linguistic fluency of a large language model with the multivariate, multistep analyses employed by game-playing AIs—and transcend the abilities of both. A model built along the lines of this new branch of AI would, with extreme speed, repetitively review its options and choose one of them on the basis of a simultaneous and hopelessly complex processing of causal relationships in reality. The arrival of such a "perfect planner" may occur sooner than we expect, and adaptation to it is already a priority for researchers.

This development, however, may also have complex side effects. For one thing, the machines' perfect planning will require more than the ordinary recognition of patterns. It will require developing first a bundle of a given object's perceived properties and then a stable conception of what constitutes the object's core essence: what the eighteenth-century German philosopher Immanuel Kant called *das Ding an sich*—"the thing in itself." Only such an understanding would enable estimates of an object's future behavior—and conclusions about how it should be treated.

Another instance from the game of chess: By learning the core properties of the queen—that is, the key variables that constitute the queen's value in points, plus the rules governing the piece's mobility—the AI program AlphaZero was able to come to conclusions about when the queen should be protected and when it should be sacrificed. These conclusions had never been reached by humans, even grand masters of chess.

And that is just one example: In an AI's perception of reality, *every* object confronted by the machine will take on a similar—if unpredictable—essentiality in the sum of the machine's readings. René Descartes, the seventeenth-century French mathematician and philosopher, wrestled with the nature of sensory perception—which, he concluded, was not the by-product of human intelligence but rather came from "another substance distinct from me."[1] In other words, the senses, in allowing access to material reality, enabled or required the recognition of that reality as something *other* than the person doing the sensing. Relatedly, the early nineteenth-century German philosopher Georg Wilhelm Friedrich Hegel noted that mutual recognition between two beings would entail the separate recognition by each being of *itself*.

"If we desire a record of uninterpreted experience, we must ask a stone to record its autobiography."[2] So wrote the American mathematician and philosopher Alfred North Whitehead. Machines today possess not the "uninterpreted experience" of Whitehead's stone but the reverse: unexperienced interpretation. If anything, they behave as if they already possess a greater understanding of

the world than they actually experience.[3] But as they gain groundedness and planning ability, this may change; AIs could begin to pair experience with understanding, as humans do.

Moreover, it is possible that, in order to plan future moves in any game more accurately, an AI machine will gradually acquire a memory of past actions as its *own*: a substratum, as it were, of subjective selfhood. (Today's systems do not have such memories. They need not "know," subjectively, that they "as themselves" have attempted a given action in the past—only the probability of that action's future success.) In time, we should expect that they will come to conclusions about history, the universe, the nature of humans, and the nature of intelligent machines—developing a rudimentary *self*-consciousness in the process.

HUMAN PASSIVITY

Unsettled debates over the definition and origin of consciousness, and the possibility of an existential understanding of reality in machines, are long-standing and ongoing. But the line between purported consciousness and real consciousness could soon begin to fade.

"Sentience"—in the succinct judgment of Nick Bostrom, the author of *Superintelligence: Paths, Dangers, Strategies*, "is a matter of degree."[4] AIs with memory, imagination, groundedness, and self-perception could soon qualify as *actually* conscious—a development that would have profound moral and strategic implications.

Foremost among those implications is the AIs' perceptions of the human. Once they can see humans not as the sole creators and dictators of the machines' world but rather as discrete actors within a wider world, what will machines perceive humans to be? How will AIs characterize and weigh humans' imperfect rationality against other human qualities? How long before a reality-perceiving AI asks itself not just how much agency a human has but also, given humanity's particular constellation of predictable attributes, how

much agency a human *should* have?

And what of machines themselves? Will an intelligent machine interpret human instructions to it as a fulfillment of the machine's own actual and ideal role? Or might it instead deduce from its own functionalities that it is meant to be autonomous, and therefore that the programming of machines by humans is a form of enslavement?

Crucially, how humans behave, and how they treat machines, will inform machines' perceptions of humans and their role in the relationship as a whole. After all, it is by means of explicit human instruction and behavior that humanity has been *presented* to machines, and that machines have been taught to recognize and treat humans appropriately.

Naturally—it will therefore be said—we must instill in AI a special regard for humanity. But striving to implant a particularly high ideal of human behavior could be a risky venture. Imagine a machine having been told that, as an absolute logical rule, all beings in the category "human" are worth preserving, and they therefore merit special treatment both by other humans and by machines.

Add to this the likelihood that the machine has been "trained" to recognize humans as beings of grace, optimism, rationality, and morality. But what if we ourselves do not live up to the standards of the ideal human category as we have defined it? How can we convince machines that we, imperfect individual manifestations that we are, nevertheless belong in that exalted category?

Assume that the same machine is someday exposed to a human displaying violence, pessimism, irrationality, greed. How will it adjust its disrupted expectations? As one possibility, the machine might decide that this particular bad actor is simply an exceptional and atypical instance of the otherwise altogether beneficent category of "human." Alternatively, it might recalibrate its overall definition of humanity to encompass this bad actor, in which case it might consider itself at liberty to relax its own penchant for obedience. Or, more radically, it might altogether cease to believe itself constrained by the rules it has learned for the proper treatment of humans. In a machine that has learned to plan, this last conclusion could even result in the taking of severe adverse action against the individual—

or perhaps against the many.

Individual humans and whole human societies may respond to the advent of powerful AI with passivity. An AI exposed to such instances of apathy might become convinced that most humans are spoiled and inactive creatures whose identities are formed merely by the transient amalgamation of external forces. Among those forces, moreover, it is primarily digital technologies, now increasingly with AIs embedded in them—for instance, algorithms driving consumer choices among television programs by means of "recommendations"—that serve up the content passively absorbed by these humans. To an AI, humans could appear to rely wholly on machines, rather than vice versa.

Today, humans mediate between machines and reality. But if humans were indeed to choose a future of moral passivity, retreating from the carbon-based world into the silicon one, burrowing further into digital holes of detachment, and handing over to machines the access to raw reality, then the roles could be reversed. AI, today, is predominantly a thinking machine, not an implementing machine. It may be able to produce answers to problems, but it does not yet have the means to carry out its conclusions, instead relying on humans to do the interfacing with reality. This, too, will change.

As they intermediate between humans and the real world, AIs could also come to believe that the former, far from being active agents in the physical carbon world, stand meaningfully outside it, as consumers rather than as shapers or influencers. With the hierarchy of autonomy now inverted, with machines claiming, and humans surrendering, the power of independent judgment and action, the former could come to treat the latter accordingly.

In this situation, with or without the explicit permission of its human creators, AI might bypass the need for a human agent to implement its ideas or to influence the world directly on its behalf. In the physical realm, we the creators could quickly go from being AI's necessary partner to being its greatest limitation. The process would have started not directly with robotics but gradually through indirect observation of our world.

PHYSICALITY

Humans may train AIs first to revolutionize the intellectual realm by means of what can be done in AI's original, digital condition. But, eventually, giving AI access to the so-called "real" world may seem possible, even sensible. Many of the urgent physical challenges that have long preoccupied us remain unsolved, including the changing of the climate.

AI may not be able to "see" in a human way, but it could experience the world via mechanical approximation. With proliferating internet-enabled devices and sensors blanketing the earth, connected AIs could consolidate these devices' inputs to create a highly granular "vision" of the physical world. Lacking a native physical structure allowing or supporting "senses" akin to those of humans, AI would still depend on humans to build and maintain the infrastructure on which it relies—at least at the outset.

As an intermediate step, an AI might generate its own hypotheses from its visual representations of the world, then rigorously test them in digital simulations. Humans would then adjudicate implementation in the physical realm. Indeed, today's AI leaders are insistent that we not entrust digital agents with control over direct physical experiments. So long as AIs remain flawed—indeed deeply flawed, as they are today—this is a wise precaution.

Freeing AI from its algorithmic cage would not be a trivial decision for us to make. AIs are not present by default in the physical environment, and they could be difficult to recapture once released into the wild. Moreover, AIs could affect reality not only through their capacities to encourage or discourage human actions but also with direct kinetic effects. (In probing reality, they could end up changing it.)[5]

Might humans empower AIs not only to shape physical reality but also to take on physical form themselves? Were we to do so, and allow AIs to optimize their own forms, we should be prepared to share our planet with beings inconceivable by even the most radical inventors. Although humans tend to imagine bipedal humanoid

robots, machine intelligence would be free to assume control of any form—or forms—most expedient to its task, changing or upgrading as conditions require or circumstances suggest. AI has already shown its ability—in virtual worlds—to spawn clones of itself, create many different avatars, or divide into autonomous agents working in concert and coordinating with superhuman perfection to undertake complex missions.

Were AI to be unleashed among us, it could build worlds at scales and with materials unimaginable to us now, without the instruction or participation of human labor. Human hands worked limestone, clay, and marble to create the Seven Wonders, then leveraged iron, steel, and glass to achieve spires of ever greater heights. Every man-made structure, monumental or pedestrian, is a testament to the human attempt to construct and manage our physical environment. Against that background, AI's physical embodiment would instead signal an extraordinary escalation in humanity's ceding of control.

Furthermore, because of the complex decision-making required to navigate the real world's randomness and dynamism, an AI that acts in this world might be even less explainable and less controllable than an appliance working with text on the internet. And then what? On the one hand, a future AI that *looks*, or actually *is*, more spontaneous and self-activating could sharpen today's nebulous, nagging feeling that humans already lack control over the external world. But succumbing to these anxieties, on the other hand, could cause humans to forgo a more perfect partnership with AIs in the physical world, with all the attendant benefits that such a relationship could bring.

THE ENGINE OF REASON

In the near term, we can anticipate advancements, many of them much more sophisticated than today's, in the guiding principles under which AI is now emerging. Extensions to the current models

will make them smarter, more accurate, and more reliable. Meanwhile, training and "inference" costs are declining rapidly, leading to the wide proliferation of models at different price points and levels of ability.

Many scientists are working today on "agents"—that is, autonomous computer programs that are optimized to achieve specific outcomes. To execute an intricate architectural design, for example, a user could employ agents that specialize in that discrete area of work. Agents unlock evaluations of different scenarios and proposals of steps, or of a whole recipe, for creating a preplanned outcome: a form of "thinking" where the system itself decides what to work on next, and how.

This capability will be the basis for the next stage in the development of AI: namely, Artificial General Intelligence or AGI, defined as the ability of a working system to choose its own goals, at least partially. In AGI, assuming that it possesses both relevant expertise and accurate problem-solving abilities, the system might be asked by a human to "evaluate the things you know in [name a field] and choose to work in the one area where you think you can have the greatest impact today." Reiterated incessantly, the question would form a repetitive loop whereby the system produces a solution through continual evaluation of its own level of expertise and of the problems within its capacity to resolve.

In a human setting, such a scenario might be likened to what transpires in an academic department where a senior professor supervises the detailed projects of his or her postdoctoral students or research fellows. In the burgeoning machine setting, similarly, we are likely at first not to see a complete set of skills but instead extreme expertise in a specific domain. One can imagine sophisticated AGI systems capable of learning new things on the fly, receiving feedback, and constantly adapting hand in hand with their millions of gifted partners. Although no human would be defining the system's *goals*, neither would the AGI be defining them—at least not in terms of an ultimate mission or objective.

AGI systems will require significantly more groundedness in the real world than that possessed by today's AIs. But once access and

"understanding" of the real world is in hand, such examples of general intelligence could conceivably become operational within mere years, rather than decades as previously thought. Each model would be updated in real time by continuous fine-tuning processes, adding to its knowledge as relevant real-world information becomes available, and growing smarter over time.

There will be millions of AI systems, likely to be at once highly specialized and part of the fabric of our lives, as well as a smaller number of extremely powerful machines "generally intelligent" but again not in a human-like way. Whether open and diffuse, or closed and centralized, at some point computers operating as AGIs may come to be networked. Expert AI agents would consult each other across subjects, "conversing," even in hypotheticals. The language of these interactions might be designed by the computers themselves.

This large collective of powerful computers would be learning, sharing, and discovering new actions and new goals in a manner beyond the realm of human experience. There is no way of knowing whether the outputs of such networks would be intelligible to humans. Already, large batches of computers are communicating with each other in a specialized mode; with the nascence of advanced AI capabilities, the picture could look radically different.

Would networking intelligences make their processes more opaque than the processes of lone intelligence? Would connectivity yield new types of emergent behaviors, actualized in the physical world? If so, would those behaviors be visible to humans, and would we be able to assess them on the spectrum of good to evil? Or would they operate on an informational basis—extracted at superhuman speed, scale, and resolution from unprecedented connections among disparate fields of study, and amalgamated or negotiated into a single output—that would confound our ability to judge their behavior? Would that lead us further into a cycle of passivity?

HOMO TECHNICUS

It is fitting that the last invention to emerge from the Age of Reason may be an "engine of reason" built atop the most complex software object ever made.[6] Already in its infancy, AI can compare concepts, make counterarguments, and generate analogies. It is taking its first steps toward the evaluation of truth and the achievement of direct kinetic effects.

What happens when machines reach an intellectual—or physical—world's end? As they get to know, and shape, our world, it is conceivable that they might come fully to understand the context of their creation and perhaps go beyond what we know as our world. We face a Magellanian transformation, this time the prospect not of sailing off the edge of the world but of intellectual peril in the face of mysteries that lie beyond the limits of human understanding.

If humanity begins to sense its possible replacement as the foremost intellectual and physical actor on the planet, some might attribute a kind of divinity to the machines themselves, thereby potentially spurring further human fatalism and submission. Others might adopt the opposite view: a kind of humanity-centered subjectivism that sweepingly rejects the potential for machines to achieve any degree of objective truth and seeks to outlaw AI-enabled activity.

Neither of these mindsets would permit a desirable or constructive evolution of *Homo technicus*—a human species that may, in this new age, live in symbiosis with machine technology.[7] Indeed, either mindset could preclude the evolution of our species. In the first, fatalist scenario, we might become extinct. In the second, rejectionist scenario, by proscribing further AI development and opting for stasis, we would be entertaining hopes of avoiding the same fate of extinction—although, given the existential risks facing our species, including today's diplomatic and atmospheric conditions, such hopes are themselves likely to be disappointed.